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Hierarchical memory-based deep reinforcement learning in simulated survival environments

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ABSTRACT

In complex decision-making tasks, agents often suffer from memory interference, inefficient exploration under sparse rewards, and limited adaptability in dynamic environments. While existing deep reinforcement learning (DRL) methods often suffer from performance limitations when handling long-horizon tasks due to memory interference, insufficient exploration, and poor adaptability. Inspired by multi-timescale memory mechanisms in neuroscience, this article proposes a hierarchical memory-based DRL (HM-DRL) architecture that integrates three complementary memory layers: a perceptual memory layer for real-time detection of environmental changes, an episodic memory layer for structured event sequence storage, and an abstract memory layer that constructs causal graph and supports backward reasoning. Furthermore, HM-DRL incorporates a dynamic gating mechanism and a compound reward function to facilitate effective integration of the policy network and memory outputs, thereby enhancing policy optimization and crisis response. Within the designed open-ended multi-task simulated survival environment, the HM-DRL architecture demonstrates substantial improvements in mitigating memory interference during long-horizon tasks, enhancing adaptability to sudden crises and dynamic environments, and improving learning efficiency under sparse reward scenarios. This article offers an effective paradigm for addressing core memory management challenges in long-horizon decision-making and establishes a foundation for developing agents with causal reasoning capabilities.

1. Introduction

Open-world environments (e.g., Minecraft) have emerged as critical testbeds for agent decision-making research, characterized by high-dimensional state spaces, dynamic environments, concurrent task demands, and long-horizon dependencies (Fan et al., 2022; Yuan et al., 2023). Compared with the notable success of deep reinforcement learning (DRL) in closed settings like Atari games and robotic control (Johannink et al., 2019; Kaiser et al., 2019; Schulman et al., 2017; Zhang et al., 2025a), DRL agents in open environments face critical limitations, particularly in memory management for long-horizon reasoning. These limitations manifest in several ways. First, conventional experience replay mechanisms often struggle in long-horizon tasks (Neves et al., 2024). Second, under sparse reward scenarios, agents face significant exploration challenges (Riedmiller et al., 2018) and often rely on additional guidance such as demonstrations (Zhang et al., 2025b). Third, the learned policies frequently exhibit poor adaptability to unexpected environment dynamics (Nagabandi et al., 2018). These challenges col-

lectively hinder effective long-horizon decision-making in open-world environments.

To address these limitations, memory-enhanced architectures have been introduced in DRL, incorporating adaptive retrieval mechanisms and structured memory representations, such as hierarchical memory transformers (He et al., 2025) and memory-augmented RL frameworks (Wang et al., 2025). Although these models improve the agent's ability to store and access large amounts of information, they still fall short in open-world settings, suffering from memory interference and stability issues (Le et al., 2024), and may lack effective mechanisms to dynamical updating memory representations. Furthermore, transformer-based architectures face challenges in long-horizon reasoning under continual interaction settings, due to the computational complexity of attention and constrained context windows (Omidi et al., 2025). As a result, existing memory extensions increase capacity but often lack explicit structured, multi-timescale memory organization needed for robust long-horizon reasoning.

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Neuroscientific evidence provides a compelling reference: human memory operates through a hierarchically layered and multi-timescale architecture (Momennejad, 2024; Vatansever et al., 2021). Specifically, the hippocampus encodes episodic memories (Schapiro et al., 2017; Tulving, 1985), enabling the rapid recording of spatiotemporally specific events, such as concrete goal-directed interactions with the environment. The neocortex abstracts generalizable knowledge across experiences, supporting the formation of semantic rules and action–outcome regularities (Badre & Nee, 2018; Ghosh & Gilboa, 2014). Meanwhile, perceptual and working-memory systems maintain a compressed, attention-focused representation of recent sensory input, allowing the agent to detect salient changes and update its internal state in real time (Cowan, 2008). These subsystems maintain distinct temporal characteristics, but long-horizon reasoning cannot be achieved through isolated processing. Effective behavior depends on coordinated information flow—from perceptual signals, to episodic events, and ultimately to abstract knowledge representations—which motivates the need for an integrated hierarchical memory design.

Building on the necessity of both hierarchical structure and coordinated interaction among memory systems, the Complementary Learning Systems (CLS) theory (Kumaran et al., 2016; McClelland et al., 1995; Schapiro et al., 2017) provides a computational account of how the hippocampus and neocortex cooperate through asynchronous consolidation mechanisms that protect long-term knowledge from being overwritten by new experiences while maintaining adaptability to novel situations. O'Reilly and Norman (2002) demonstrate that, although functionally distinct, these systems exhibit strong complementarity within the memory architecture. While some prior DRL methods have attempted to simulate hippocampal functions (Ahn, 2025; Eichenbaum, 2015; Hu et al., 2021; Li et al., 2025), they typically lack explicit mechanisms for memory separation and real-time coordination across temporal scales. As a result, DRL agents often suffer from policy oscillation and inefficient exploration in complex, dynamic environments. In addition, Pan et al. (2025) show that positive emotional signals can enhance memory and increase the salience of experiences by promoting memory reinstatement during repeated learning.

Motivated by these neuroscientific insights and the limitations of existing DRL memory designs, we propose a novel Hierarchical Memory-Based Deep Reinforcement Learning (HM-DRL) architecture. The main contributions of this article can be summarized as follows:

- 1) A neuroscience-inspired three-layer memory architecture, consisting of:
 - (a) Perceptual memory layer: continuously monitors environmental changes and detects critical state shifts;
 - (b) Episodic memory layer: stores temporally structured event sequences, enabling fast retrieval and experience matching;
 - (c) Abstract memory layer: encodes precondition–action–effect rules from experience and constructs a structured causal graph to support planning via backward reasoning mechanism.
- 2) A dynamic gating mechanism driven by environmental features that adaptively fuses policy outputs and memory-guided suggestions via learned weighting coefficients, enhancing behavioral flexibility and environmental adaptability.
- 3) A compound reward function that integrates external environmental rewards with multiple intrinsic motivations, including novelty-driven exploration, prediction-consistency rewards, and path-following incentives, enabling the agent to sustain effective learning under sparse reward scenarios.

2. Related work

2.1. Modeling and evolution of memory mechanisms in DRL

Memory plays a vital role in advancing reinforcement learning, especially in tasks requiring long-horizon dependencies and struc-

tured recall. One of the earliest memory mechanisms, the experience replay strategy used in Deep Q-Networks (DQN) (Mnih et al., 2015), stores past state-action-reward sequences in a buffer to break sample correlations and improve training stability. However, this method primarily relies on unstructured storage and passive sampling, and is incapable of active retrieval or context-aware policy adaptation.

To better encode and leverage historical task information, researchers explore more structured memory modeling paradigms. Banino et al. (2020) propose the MEMO model, which separates memory components to support long-horizon reasoning, demonstrating effectiveness in complex cognitive tasks. However, its architecture remains constrained by relatively fixed memory channels. Subsequent studies have explored more adaptive retrieval mechanisms. Hu et al. (2021) introduce a generalizable episodic memory framework for deep reinforcement learning, showing that episodic retrieval can improve policy learning by enabling agents to recall useful past experiences across tasks. More recently, Jin et al. (2023) propose a task-adaptive retrieval mechanism via hypernetworks, allowing the agent to dynamically modulate memory access according to task context.

In parallel, Kim et al. (2023) propose a multi-component architecture integrating short-term, episodic, and semantic memory modules, highlighting the potential of memory collaboration for more flexible behavior generation. While these methods improve retrieval and adaptability, Omid et al. (2025) note that existing methods primarily focus on specific aspects such as modular design or task-adaptive retrieval, with limited exploration of structured memory organization for long-horizon reasoning.

2.2. Cognitive architectures and neuroscience-inspired memory models

Neuroscience provides a rich foundation for designing multi-timescale memory systems. The hippocampus rapidly encodes episodic traces and replays past trajectories, enabling fast behavioral adaptation (Singh et al., 2022; Whelan et al., 2022; Yu et al., 2025). The neocortex gradually extracts abstract structure and semantic knowledge across experiences, supporting generalization and long-horizon planning. This division of labor inspires hierarchical memory architectures in artificial agents. From a computational viewpoint, Botvinick et al. (2009) demonstrate that hierarchical RL aligns well with layered cognitive control observed in humans and animals. Ororbia and Kelly (2022) further extend these ideas through the CogNGen cognitive architecture.

RL systems inspired by hippocampal mechanisms have incorporated episodic retrieval and memory-guided decision processes. Retrieval-based methods enable agents to recall past experiences to guide current decisions and improve sample efficiency (Hu et al., 2021; Liang et al., 2024). More recent methods further introduce adaptive memory utilization mechanisms, allowing agents to dynamically select and refine useful experiences based on task context (Jin et al., 2023; Zhang et al., 2026). These methods demonstrate that principled memory design can improve generalization and learning efficiency. However, existing methods often model episodic retrieval and higher-level abstraction as separate components, with fewer efforts devoted to their coordinated interaction across temporal scales.

A separate line of work studies causal RL, where causal graphs are used for policy evaluation, counterfactual reasoning, or modeling environment-level intervention structures (Zeng et al., 2025). These methods typically focus on learning relatively stable causal relations at the level of environment dynamics. In contrast, memory-based agents emphasize how causal abstractions can be formed incrementally from episodic interactions and adapted over time to support planning. Cognitive studies similarly suggest that humans construct causal schemas from experience and apply them flexibly during decision-making (Lake et al., 2017). Accordingly, causal RLs and memory-based methods reflect different modeling perspectives within sequential decision-making.

Table 1
Summary of key notations used in HM-DRL.

Symbol	Description
s_t, s_{t-1}	Raw environmental states at time t and $t-1$
$s_{t,\text{loc}}$	Location-related subvector of s_t
n	Dimensionality of raw state vector s_t
M^p, M^e, M^a	Perceptual, episodic, and abstract memory layers
r_{ext}	Extrinsic reward from the environment
r_{novel}	Intrinsic novelty reward (location unfamiliarity)
r_{predict}	Intrinsic prediction-consistency reward (episodic)
r_{path}	Path-following reward from planning
λ_{novel}	Scaling coefficient for r_{novel}
λ_{pred}	Scaling coefficient for r_{predict}
λ_{path}	Scaling coefficient for r_{path} (linearly decayed)
τ_{act}	Activation threshold for state change detection
τ_{unc}	Uncertainty threshold for novelty activation
τ_{sim}	Similarity threshold for storing a new episodic event
τ_{err}	The prediction error threshold r_{predict}
$\text{sim}(x, y)$	Cosine similarity: $\text{sim}(x, y) = \frac{x \cdot y}{\ x\ \ y\ }$
unc_t	Location uncertainty at time t (Eq. (5))
CAF_t	Crisis Assessment Function value at time t (Eq. (7))
N	Number of critical resource types used in CAF
r_t^j	Current quantity of the j th resource at time t
T_j	Crisis threshold of the j th resource
w_j	Weight of the j th resource in CAF
Γ_a	Preconditions required to execute action a
Δ_a	Effects produced by action a
Path	Planned action sequence from backward reasoning
β	Maximum depth of backward search in planning
$\alpha \in [0, 1]$	Gating weight produced by $g_\theta(\cdot)$

3. Methods

To facilitate readability, Table 1 summarizes the key notations used throughout this section.

3.1. Hierarchical memory design

The overall architecture of HM-DRL is illustrated in Fig. 1, with its core operational procedure comprising four stages: 1) the perceptual memory layer continuously acquires environmental state vectors, monitoring critical feature dimensions (e.g., weather changes, danger signals); 2) the processed state is forwarded to a PPO-based policy network for action selection; 3) upon action execution and reward acquisition, the resulting transition is forwarded to the episodic memory layer. This layer selectively stores state-action-transition tuples only when the extrinsic reward satisfies $r_{\text{ext}} \geq 0$. To preserve diversity and avoid redundant storage, the current state is compared with previously stored states associated with the same action using cosine similarity, and a new episodic event is stored only if it is sufficiently dissimilar; and 4) the abstract memory layer periodically analyzes characteristic patterns of high-frequency events within the episodic memory layer, parsing critical state information (e.g., location, resources, weather). Through computational quantification of state deltas, it constructs structural causal graph of premise-action-effect. When abrupt environmental shifts (e.g., hazard triggers) are detected by the perceptual memory layer, the causal graph derive action sequences to resolve emergent states. These sequences are forwarded to the agent alongside gating network outputs. The gating network generates a scalar fusion weight based on real-time environmental states, enabling the agent to determine action-selection trustworthiness.

3.1.1. Perceptual memory layer

The perceptual memory layer M^p is designed to simulate selected functions of short-term working memory, specifically short-horizon change detection. It maintains a compact representation of recent environmental changes, enabling real-time encoding of the current state vector s_t and the identification of salient environmental changes. As a preprocessing step within the perceptual module, all state dimensions

are normalized to the $[0, 1]$ range. For each state dimension i , the change magnitude, attention weight, and weighted change intensity are computed as:

$$c_i = |s_{t,i} - s_{t-1,i}|, \quad i = 1, 2, \dots, n, \quad (1)$$

$$w_i = \frac{e^{c_i}}{\sum_{j=1}^n e^{c_j}}, \quad (2)$$

$$u_i = c_i \cdot w_i, \quad (3)$$

where c_i measures the instantaneous change in the i th state dimension, and w_i provides an attention allocation proportional to the relative magnitude of that change.

A state transition in dimension i is formally defined to occur when the following activation condition holds:

$$\mathcal{A}_i = \mathbf{I}\left(c_i \geq \tau_{\text{act}} \wedge u_i > 0.05 \cdot \max_{\text{history}}(s_i)\right), \quad (4)$$

where $\mathbf{I}(\cdot)$ is the indicator function, τ_{act} is activation threshold for state change detection, and $\max_{\text{history}}(s_i)$ denotes the maximum value observed for dimension i within the current episode. Thus, $\mathcal{A}_i = 1$ denotes that a significant change in the i th state dimension satisfies the attention-weighted activation condition, and a global transition is detected whenever $\exists i$ such that $\mathcal{A}_i = 1$. High-priority dimensions (e.g., food, resource stability) directly trigger the dynamic gating mechanism once activated.

To regulate memory usage based on environmental familiarity, the system estimates how familiar the current location is relative to previously visited locations stored in episodic memory. Specifically, the uncertainty unc_t is defined using the maximum cosine similarity between the current location feature $s_{t,\text{loc}}$ and all stored location vectors M_{loc}^e :

$$\text{unc}_t = 1 - \max_{l_k \in M_{\text{loc}}^e} \text{sim}(s_{t,\text{loc}}, l_k), \quad (5)$$

where $s_{t,\text{loc}}$ denotes the location-related subvector of s_t , and l_k denotes a stored location vector in the episodic memory. A larger unc_t indicates lower environmental familiarity. Based on this measure, M^p issues a novelty intrinsic reward:

$$r_{\text{novel}} = \lambda_{\text{novel}} \cdot \mathbf{I}(\text{unc}_t \geq \tau_{\text{unc}}), \quad (6)$$

where τ_{unc} denotes a location uncertainty threshold.

Meanwhile, we define a Crisis Assessment Function (CAF) to evaluate the severity of crisis conditions, which is as follows:

$$\text{CAF}_t = \sum_{j=1}^N w_j \cdot \max\left(0, \frac{T_j - r_t^j}{T_j}\right), \quad (7)$$

where N denotes the number of critical resource types, r_t^j is the current quantity of the j th resource at time t , T_j is the crisis threshold for the j th resource, and w_j is its corresponding weight. The resulting CAF_t is forwarded to the dynamic gating mechanism and backward reasoning mechanism to determine whether crisis intervention should be triggered. Thus, M^p serves as a working-memory controller that triggers episodic writing, novelty signalling, and crisis-aware gating.

3.1.2. Episodic memory layer

The episodic memory layer M^e stores only those events that are considered successful or informative for learning. Specifically, an event is eligible for storage only when the extrinsic reward satisfies $r_{\text{ext}} \geq 0$, preventing failed or noisy transitions from contaminating the long-sequence causal structure.

When the agent executes action a_{t-1} in state s_{t-1} and receives a non-negative reward, the memory module retrieves all previously stored events associated with the same action:

$$S_{a_{t-1}} = \{s_k \mid a_k = a_{t-1}\}. \quad (8)$$

To maintain diversity and avoid storing redundant experiences, the cosine similarity between s_{t-1} and every state in $S_{a_{t-1}}$ is evaluated:

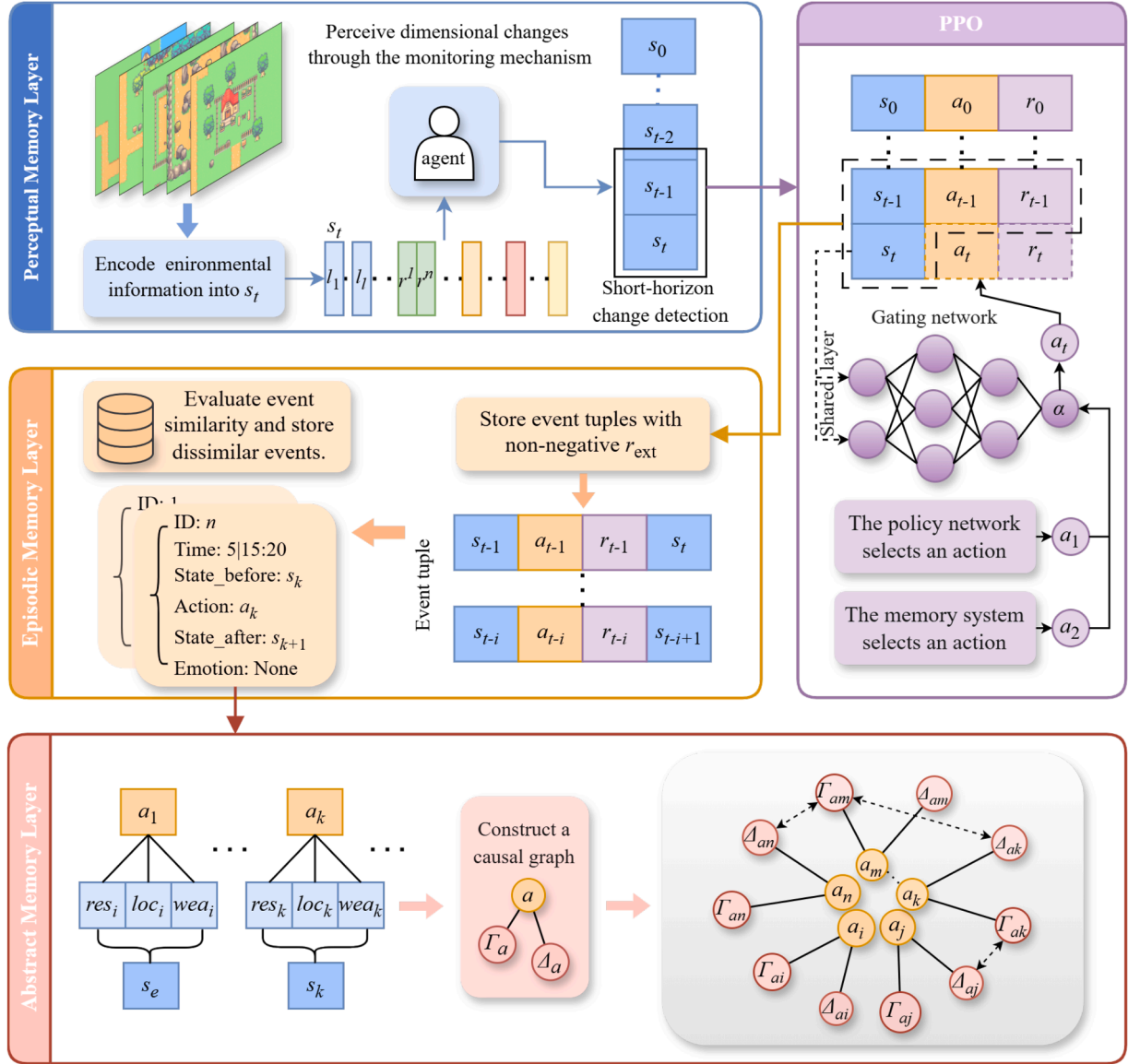


Fig. 1. Architecture of HM-DRL. HM-DRL consists of a perception memory layer, episodic memory layer, and abstract memory layer. Bidirectional dashed arrows in the causal graph indicate that the outcome of one action may serve as the prerequisite for another, although they are not strictly equivalent.

$\text{sim}(s_{t-1}, s_k)$. A new event is written to memory only if it differs sufficiently from all previously stored events associated with the same action:

$$\max_{s_k \in S_{a_{t-1}}} \text{sim}(s_{t-1}, s_k) < \tau_{\text{sim}}, \quad (9)$$

where τ_{sim} is a similarity threshold for storing an episodic event. This action-conditioned storage rule ensures that M^e contains diverse, non-failed, and behaviorally meaningful experiences.

Each stored event is formatted as:

$$\langle id, time, s_k, a_k, s_{k+1}, emotion \rangle, \quad (10)$$

where *emotion* is an emotional signal associated with each episodic event to capture its salience or preference for the agent, which can be instantiated as a scalar value reflecting the relative importance of the experience, and may be used to modulate memory retrieval or prioritization. The memory is implemented as a bounded double-ended queue, preventing unbounded growth and maintaining a manageable retrieval cost.

Before executing action a_t in state s_t , the agent retrieves all historical states in which the same action was previously executed:

$$S_{a_t} = \{ s_k \mid a_k = a_t \}. \quad (11)$$

Based on these candidates, the episodic layer provides a prediction-consistency intrinsic reward:

$$r_{\text{predict}} = \lambda_{\text{pred}} \cdot \mathbf{I} \left(\min_{s_k \in S_{a_t}} \|s_t - s_k\|^2 < \tau_{\text{err}} \right), \quad (12)$$

where λ_{pred} is a constant reward coefficient and τ_{err} is the prediction error threshold. This term encourages the agent to exploit previously observed consistent transitions in similar states.

Through this action-conditioned and diversity-preserving mechanism, M^e captures a compact but expressive set of successful episodic traces, enabling downstream backward reasoning while avoiding the propagation of failed or misleading experiences into the causal graph.

3.1.3. Abstract memory layer

The abstract memory layer M^a , inspired by the slow-learning mechanism of the neocortex, serves as the agent's knowledge base. By an-

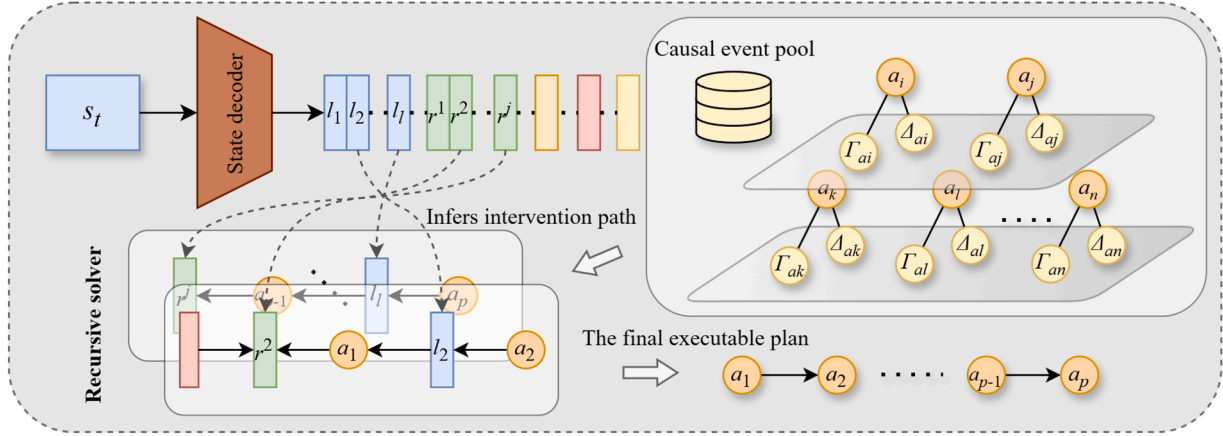


Fig. 2. Crisis resolution process CG-based backward reasoning. In the processed environmental state vector, blue represents location dimensions, green denotes resource dimensions, orange indicates weather status, red marks critical monitored dimensions, and yellow refers to other dimensions. The system detects significant changes in critical dimensions and infers intervention path by combining resource and location information with the CG.

alyzing specific event tuples $\langle s_k, a_k, s_{k+1} \rangle$ stored in the M^e , the system processes environmental states, extracts generalizable knowledge, and constructs or updates a Causal Graph (CG) composed of abstract precondition–action–effect knowledge.

Empirical analysis across multiple tasks shows that changes in key state dimensions often reveal stable action–environment causal relationships. Building on this insight, we adopt a state-differencing-based mechanism to extract action preconditions and effects. Specifically, the state change vector $\delta_a = s_{k+1} - s_k$ is first computed, and significant changes in position or resource-related dimensions are identified. These changes are then abstracted into a symbolic effect set Δ_a , while the corresponding contextual requirements are summarized as the precondition set Γ_a . The extraction process identifies state attributes whose changes are causally attributable to action a , without assuming any fixed sign pattern in the numerical change δ_a .

Eventually, a canonical set of ternary rule tuples $\langle \Gamma_a, a, \Delta_a \rangle$ is formulated, where only distinct and non-redundant causal effects are retained. In the CG, nodes represent states or resources, and edges denote state transitions caused by actions. If state or resource nodes across different events share similar conditions, associative edges across actions are established, forming a densely connected CG.

To support decision-making and crisis intervention based on the CG, we design a backward reasoning mechanism that, given the current state, generates a minimal action sequence sufficient to satisfy the required goal conditions. To ensure reasoning efficiency, both loop detection and depth-limiting constraints are applied, thereby avoiding invalid or redundant exploration. The overall process is illustrated in Fig. 2.

The backward reasoning mechanism consists of the following components:

- 1) State decoder: this module maps s_t into a discretized state vector $s_t = (l_t, \dots, r_t^1, r_t^2, \dots, r_t^j, \dots)$, where $l_t \in L$ denotes the agent's current location, and $r_t^j \in R$ represents the quantity of the j th resource.
- 2) Causal event pool E : the set of ternary rule tuples $E = \langle \Gamma_a, a, \Delta_a \rangle$ derived from the CG.
- 3) Recursive solver: a depth-limited backward search algorithm that constructs intervention path. The maximum search depth is constrained by a hyperparameter β . The pseudocode of CG-based backward reasoning is illustrated in Algorithm 1.

We define the set of goal literals g that can be directly achieved by the effect Δ_a as

$$\text{Achieved}(\Delta_a, \text{goal}) = \{ g \in \text{goal} \mid \Delta_a \text{ makes } g \text{ satisfied} \}, \quad (13)$$

and denote

$$\text{Helps}(\Delta_a, \text{goal}) = (\text{goal} \cap \text{Achieved}(\Delta_a, \text{goal}) \neq \emptyset). \quad (14)$$

Algorithm 1 Backward reasoning of intervention path.

Require: Current state s_t , goal condition set G , causal event pool E , maximum depth β

Ensure: Action sequence Path (or **FAIL** if not found)

```

1: visited  $\leftarrow \emptyset$ 
2: function REGRESS(goal, depth)
3:   if  $s_t \models \text{goal}$  then
4:     return []
5:   end if
6:   if  $\text{depth} \leq 0$  or  $\text{goal} \in \text{visited}$  then
7:     return FAIL
8:   end if
9:   visited  $\leftarrow \text{visited} \cup \{\text{goal}\}$ 
10:  for each rule  $\langle \Gamma_a, a, \Delta_a \rangle \in E$  (prioritized by effect magnitude) do
11:    if  $\text{Helps}(\Delta_a, \text{goal})$  then
12:       $\text{goal}' \leftarrow (\text{goal} \setminus \text{Achieved}(\Delta_a, \text{goal})) \cup \Gamma_a$ 
13:       $\text{subplan} \leftarrow \text{REGRESS}(\text{goal}', \text{depth} - 1)$ 
14:      if  $\text{subplan} \neq \text{FAIL}$  then
15:        return  $\text{subplan} + [a]$ 
16:      end if
17:    end if
18:  end for
19:  return FAIL
20: end function
21: Upon crisis detection:
22: Path  $\leftarrow \text{REGRESS}(G, \beta)$ 
23: return Path

```

Actions are discovered in reverse chronological order during backward search, and the final executable plan is:

$$\text{Path} = (a_1, a_2, \dots, a_p). \quad (15)$$

For long-horizon tasks, we incorporate prior knowledge by embedding a teacher policy into selected causal rules during early training. During crisis conditions or long-horizon tasks, a path-following intrinsic reward r_{path} is applied:

$$r_{\text{path}} = \lambda_{\text{path}} \cdot \mathbb{I}(a_t = \text{Path}[0]), \quad (16)$$

where $\text{Path}[0]$ denotes the first action in the planned sequence, and $\lambda_{\text{path}} \in [0, 1]$ decays linearly during training, enabling the agent to progressively rely on its learned policy rather than external guidance.

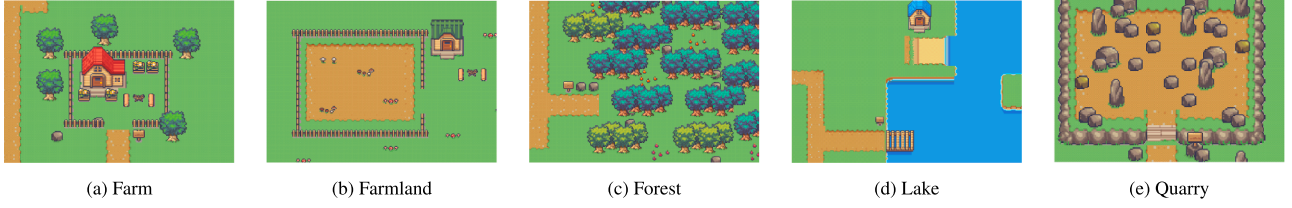


Fig. 3. Experimental environments. (a) contains a wooden house where the agent can rest and supports building workbenches for subsequent maintenance and upgrading of tools such as fishing rods. Within (b), the agent can perform actions including planting, watering, and harvesting crops to obtain food necessary for survival. (c) enables the agent to chop trees to collect wood resources. (d) allows fishing to acquire additional food sources, while (e) permits stone mining to provide raw materials for equipment upgrades.

3.2. Dynamic gating mechanism

To enhance the agent's adaptability and behavioral flexibility in complex environments, we introduced a dynamic gating mechanism. This mechanism adaptively integrates the fusion weights of policy network $\pi_\phi(a|s)$ and memory system $M(a|s)$ via a gating network $g_\theta(s)$, which dynamically adjusts the degree of influence from each component during action selection, as follows:

$$\alpha = g_\theta(h_s) \in [0, 1], \quad (17)$$

$$P_{\text{final}}(a|s) = \alpha \cdot P_\pi(a|s) + (1 - \alpha) \cdot P_M(a|s), \quad (18)$$

where h_s denotes the feature vector generated by the shared feature extractor, and $P_\pi(a|s)$ and $P_M(a|s)$ denote the action distributions produced by the policy network and memory system, respectively. The structure of the gating network is defined as follows:

$$g_\theta(h_s) = \sigma(W_2 \text{ReLU}(W_1 h_s)), \quad (19)$$

where W_1 and W_2 are trainable weight matrices, and σ denotes the sigmoid activation function.

The gating network's parameters are updated end-to-end through the RL loss, allowing the gating mechanism to automatically learn optimal fusion weights during training. The resulting scalar output, α , influences the final action selection. Through this gating network output, the agent can dynamically adjust its reliance on memory-based action recommendations, determining when to prioritize memory guidance and when to suppress or disregard it.

3.3. Compound reward function and policy network

The total reward is the sum of extrinsic and three intrinsic rewards:

$$r = r_{\text{ext}} + r_{\text{novel}} + r_{\text{predict}} + r_{\text{path}}. \quad (20)$$

HM-DRL is built directly on proximal policy optimization (PPO), preserving its core architecture while introducing only two additions: 1) the dynamic gating network above, and 2) injection of the three intrinsic rewards into the reward signal. Algorithm 2 shows the algorithm flow of HM-DRL.

4. Experiments

4.1. Experimental environment

In this section, we validate the HM-DRL architecture within a simulated survival environment inspired by survival game mechanics and composed of diverse interactive elements. The environment is designed to incorporate sufficient task complexity and dynamic disturbance mechanisms while including essential modules such as resource management, task-chain dependencies, sudden crises, and dynamic weather systems. We intentionally omitted modeling the actual time delay associated with action execution to allow each decision step to focus on validating the core memory mechanisms without extending the training

Algorithm 2 HM-DRL training procedure.

Initialize: M^p , M^e , M^a , PPO policy π_ϕ , replay buffer B

- 1: **for** each episode **do**
- 2: **for** $step=1,2,\dots$ **do**
- 3: Receive state s_t from environment
- 4: Compute change detection signal via Eqs. (1)–(4)
- 5: Compute unc_t and CAF_t
- 6: Obtain a reward r_{novel} via Eq. (6)
- 7: $r_{\text{path}} \leftarrow 0$, $r_{\text{predict}} \leftarrow 0$, $\text{Path} \leftarrow \emptyset$
- 8: **if** $\mathcal{A}_t = 1$ **and** $CAF_t > 0$ **then**
- 9: $\text{Path} \leftarrow \text{PLAN}(s_t)$ (Algorithm 1)
- 10: **end if**
- 11: Select a_t via Eqs. (17)–(19)
- 12: **if** $\text{Path} \neq \emptyset$ **and** $a_t = \text{Path}[0]$ **then**
- 13: $r_{\text{path}} \leftarrow \lambda_{\text{path}}$ via Eq. (16)
- 14: **end if**
- 15: Obtain a reward r_{predict} via Eq. (12)
- 16: Execute a_t , observe r_{ext} , s_{t+1}
- 17: $r_t \leftarrow r_{\text{ext}} + r_{\text{novel}} + r_{\text{predict}} + r_{\text{path}}$
- 18: Store transition (s_t, a_t, r_t, s_{t+1}) in B
- 19: **if** $r_{\text{ext}} \geq 0$ **then**
- 20: Update M^e and M^a
- 21: **end if**
- 22: Update π_ϕ using PPO on minibatches from B
- 23: **end for**
- 24: **end for**

time, and to comprehensively assess the robustness and generalizability of HM-DRL in open-world tasks.

We first constructed a virtual environment comprising five functional regions, with the map designed using the Tiled toolkit, as shown in Fig. 3. The environment includes twelve preset scenarios. The core modules consist of a dynamic weather system, a crop cultivation system (simplified such that crop growth from planting to maturity takes approximately three days), a crisis event generator, a tool crafting and upgrade system, and fundamental survival rules including mandatory daily rest, thereby simulating the complexity and randomness inherent to open-world tasks.

The dynamic weather system encompassing conditions such as sunny, rainy, windy, and foggy weather. Rain increases soil moisture and raises stamina consumption during movement; strong winds significantly raise the consumption of outdoor actions; fog reduces the success rate of fishing activities. The system further incorporates various crisis conditions, including tool breakage and sudden food loss. Although the agent's actions are not restricted by spatial location, rewards are only granted when the action corresponds appropriately to the current scene. Executing actions in irrelevant regions yields no reward, consumes stamina and time, and incurs a small penalty. To enhance task challenge and rationality, several rule constraints are introduced: failure to rest on time, exhaustion of stamina, or excessive hunger immediately terminates the episode with a penalty. While successful food-

Table 2
Indirect resource dependencies in survival environment.

Resource	Direct impact	Affects food production	Affects crisis recovery	Affects long-term survival
Food	Yes	Yes	Yes	Yes
Wood	No	Repairs tools	Yes	Yes
Iron	No	Upgrades tools	Yes	Yes
Rod durability	No	Limits fishing attempts	Yes	Yes
Growth stage	No	Determines crop yield	Yes	Yes
Soil moisture	No	Affects crop growth rate	Yes	Yes
Tool platform	No	Enables crafting	Yes	Yes
Weather state	No	Influences harvest, fishing efficiency, and stamina	Yes	Yes

related actions are limited to five per day and daily idle time is limited to five hours, with penalties applied when either limit is exceeded. The agent's primary optimization objective is to maximize survival duration and overall score. To prevent policy stagnation in local optima, we also introduced an action diversity constraint. The overall experimental design adheres to survival principles.

Although food is the only variable that directly determines survival in the environment, the agent's survival depends on multiple interacting resources that indirectly influence food acquisition and recovery from crises (e.g., tool durability, crop growth, and weather). As summarized in Table 2, these variables do not always trigger failure directly, but they affect food acquisition and crisis recovery.

4.2. State and action space

A detailed description of the state representation is given in Table 3, which provides the full definition of the 28-dimensional state vector used in the experiments, including location encoding, survival metrics, resource variables, crop-growth attributes, and weather conditions.

Regarding the action space, the environment includes nineteen discrete base actions, including but not limited to movement, idling, chopping, planting, fishing, tool maintenance, and tool upgrading, among others. The constructed curriculum-style survival environment is encapsulated as a module compliant with the Gym interface specification.

4.3. Core functionality verification

All experiments were conducted under standardized hardware conditions. We incorporated a curriculum learning mechanism, dividing the training process into four progressive stages with incrementally escalating survival difficulty. Each episode was limited to 10 in-game days and at most 300 decision steps, encouraging agents to develop efficient policies within a constrained horizon. In addition, all experimental groups received instructional guidance from teacher policy. The baseline DRL algorithm for all experimental groups utilized identical PPO implementations, with hyperparameters and network architectures summarized in Table 4. The complete architecture comprises four subnetworks: a shared feature extractor, an actor network, a critic network, and a gating network. Following standard practice, only the configurations of the shared-layer and actor networks are explicitly reported, while the remaining networks adopt similar structures. The remaining networks have similar structures and are not reiterated here. Considering the longest causal chain in the experimental environment, we set $\beta = 10$. The absolute change thresholds for actions $\tau_{act} = 0.95$, $\tau_{err} = 0.05$, $\tau_{unc} = 0.9$, and $\tau_{sim} = 0.9$.

To evaluate the proposed method, we selected three memory-augmented reinforcement learning algorithms as baseline models: PPO (Schulman et al., 2017), generalizable episodic memory (GEM) (Hu et al., 2021), and episodic memory based on hypernetwork (EM-Hypernet) (Jin et al., 2023). For GEM, we retained its core mechanisms, including parameterized generalized memory, twin back-propagation networks, implicit memory-based planning, and conservative estimation mechanism. For EM-Hypernet, we preserved its key components such

Table 3
State representation used in the experiments. The full state vector has 28 dimensions.

Index	Symbol	Description
Location (5D, one-hot)		
0	l_{cabin}	At cabin
1	l_{field}	At crop field
2	l_{forest}	At forest
3	l_{lake}	At lake
4	l_{mine}	At mine
Survival indicators (5D)		
5	day_t	Day index (normalized by /20)
6	$time_t$	Current time of day (normalized by /1080)
7	$night_t$	Indicator of nighttime ($\mathbb{I}[t \geq 18 : 00]$)
8	$stamina_t$	Stamina level (/100)
9	$hunger_t$	Hunger level (/100)
Resource state (10D)		
10	food	Food amount (/20)
11	seeds	Seed amount (/20)
12	wood	Wood amount (/20)
13	rod_dur	Fishing rod durability (/20)
14	iron	Iron ore amount (/30)
15	food_low	Food shortage indicator
16	seed_low	Seed shortage indicator
17	wood_low	Wood shortage indicator
18	good_rod	Whether the agent owns the advanced rod
19	tool_platform	Whether a crafting platform is available
Crop system (4D)		
20	planted	Whether a crop is planted
21	growth	Crop growth stage (/3)
22	mature	Whether the crop is mature
23	moisture	Soil moisture (/100)
Weather (4D, one-hot)		
24	w_{sunny}	Sunny
25	w_{rain}	Rainy
26	w_{wind}	Windy
27	w_{fog}	Foggy

as task-conditioned hypernetwork, dynamic modification mechanisms, and retrieval network with attention, while replacing the CNN module with a direct use of the raw state vector. An additional objective of maximizing survival days was introduced to encourage long-horizon planning, with increased reward for longer survival. Baseline groups were also provided with teacher-guided strategies for long-horizon tasks. Teacher guidance is implemented as the r_{path} (Eq. (16)), which is applied identically to all methods and linearly decayed to zero during training. When the agent's chosen action aligned with the recommended path, a path-following reward was given.

All models were trained and evaluated under the same four-stage curriculum learning setup, and each experimental group was trained using five random seeds. In the initial training stages, food resources were intentionally provided to facilitate the learning of basic survival strategies. In the later stages, food provisioning was removed to encourage the agent to develop adaptive strategies under resource scarcity and to mitigate potential policy bias arising from reliance on the memory

Table 4
PPO configuration and network architecture.

Category	Component	Input → output dimension	Activation / Value
Hyperparameters	Batch size	–	256
	Discount factor	–	0.98
	Learning rate	–	2.5×10^{-4}
	GAE lambda	–	0.95
	PPO clipping epsilon	–	0.25
Shared network	Layer 1	28 → 256	ReLU
	Layer 2	256 → 128	ReLU
	Layer 3	128 → 128	LayerNorm
Actor network	Layer 1	128 → 64	ReLU
	Layer 2	64 → 19	Softmax

architecture, with detailed experimental configurations summarized in Table 5.

Four sets of controlled experiments were conducted with specific objectives: 1) a comparison between Experiment Group 1 and 2 to assess the effect of varying initial training durations on policy learning efficiency; 2) a comparison between Experiment Group 1 and 3 to evaluate the impact of the dynamic weather system on policy generalization; 3) Group 4 was designed to test the agent’s adaptability in response to sudden crisis events, where there was a 10% probability of critical events occurring per episode, such as abrupt depletion of core food resources or unexpected tool malfunctions.

4.4. Ablation and sensitivity study

We evaluate the necessity and robustness of the proposed HM-DRL framework through a combined ablation and sensitivity analysis. All experiments are conducted under the same environment configuration as Group 4 from the previous subsection, with dynamic weather conditions enabled, while keeping all other settings unchanged.

Ablation settings. We conduct ablation experiments by selectively removing specific components to examine their contributions to policy learning and decision-making. In particular, we consider the following four settings: 1) removing the M^a while retaining the M^p and M^e ; 2) injecting misleading action recommendation sequences with a probability of 0.1 to assess robustness against abnormal causal guidance; 3) removing prior knowledge for long-horizon tasks while preserving the complete hierarchical memory architecture; 4) retaining the full three-layer memory structure. Each ablation setting is trained using five different random seeds under identical hyperparameter configurations, and performance statistics are aggregated across seeds.

Crisis robustness settings. To assess adaptability under unexpected environmental disruptions, we further conduct 10 independent evaluation runs for each model trained with five random seeds. In each run, all agents are initially provided with three food items on the first day, after which all food resources are instantaneously removed at the beginning of the second day. This setup evaluates crisis-response behavior across different ablation settings.

Sensitivity settings. Although several thresholds in HM-DRL can be inferred from the structure of the discrete state space, some key hyperparameters still involve design choices and may influence learning behavior. To examine robustness with respect to such design-sensitive hyperparameters, we analyze the effects of two representative threshold parameters: τ_{sim} and τ_{err} . In our experiments, the values of τ_{sim} are varied in {0.7, 0.8, 0.9}, while the values of τ_{err} are varied in {0.05, 0.15, 0.25}. Each sensitivity configuration is trained using three random seeds, and results are aggregated across seeds.

5. Results

5.1. Training results

We analyze training performance as initial food resources are progressively reduced across curriculum stages. At the beginning of each stage, average rewards temporarily decline due to environmental changes requiring policy adaptation. In early training, HM-DRL and the baseline PPO exhibit similar performance, as both remain in random exploration and have not yet formed an effective CG, resulting in no observable policy advantage.

As shown in Fig. 4, although the HM-DRL modules slightly reduces cumulative rewards during early training, the per-episode average cumulative reward improves significantly as training progresses. During mid-to-late training, HM-DRL surpasses the baselines in average per-episode rewards, which is attributed to the progressive completion of the CG constructed by M^a . When facing sparse long-horizon tasks or sudden environmental changes, the agent applies a CG-based backward reasoning mechanism to generate more effective strategies, reducing ineffective exploration and improving survivability.

In the second experimental group, the early training constraint of a maximum survival duration of one day prevents the agent from capturing long-horizon delayed reward structures, leading to sub-optimal initial policy learning. By contrast, the other three groups, trained with a maximum survival duration of five days, enable learning of long-horizon dependent tasks such as periodic watering and harvesting at crop maturity.

Overall, training curves show that HM-DRL maintains high average rewards as resource levels decrease by proactively constructing CG, identifying resource acquisition paths, and executing timely interventions. In contrast, baseline agents achieve relatively high rewards only in early stages and fail to exhibit significant reward escalation in mid-to-late training, indicating that algorithms without explicit long-term goal modeling and causal reasoning are less capable of forward-looking strategy generation in deteriorating environments. These results provide further evidence for the advantage of the proposed architecture in long-horizon planning under open-ended environments.

5.2. Testing results

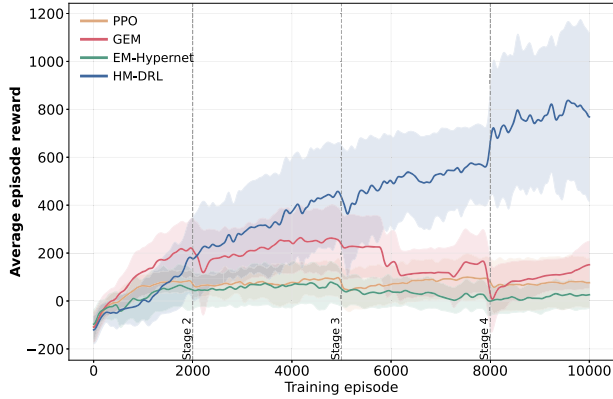
For evaluation, we conducted ten independent test episodes for each model trained with five random seeds. The testing results are visualized in Fig. 5, which provides an intuitive comparison of survival duration and accumulated rewards across different methods. The corresponding numerical statistics, reported as mean $\pm$ standard deviation over five seeds, are summarized in Table 6. Paired t-tests on per-seed average performance show that HM-DRL achieves statistically significant improvements in several task settings (e.g., Group 1 and partially Group 4), while the differences are not significant in other settings due to large variance across random seeds.

In the comparison experiment with the baseline PPO, we observed distinct behavioral differences during evaluation. PPO-trained agents maintained relatively stable exploration behavior, enabling them to consistently execute basic survival actions and achieve moderate survival duration. In contrast, agents under the EM-Hypernet condition exhibited lower performance than PPO in survival duration metric. Despite the introduction of an additional objective of maximizing survival days during training, EM-Hypernet remains biased toward task-conditioned retrieval and sequence-level optimization. In the survival environment, where task boundaries are implicit and long-term resource dynamics dominate, this inductive bias can suppress necessary exploration and hinder the acquisition of robust survival behaviors, resulting in lower overall performance than PPO.

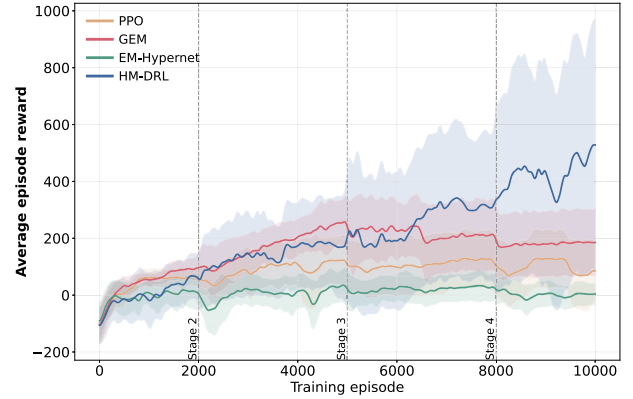
Although the twin back-propagation networks design of GEM aims to reduce overestimation in trajectory planning, it may underestimate the value of necessary actions in sparse reward environments. For instance,

Table 5
Details of experimental configurations.

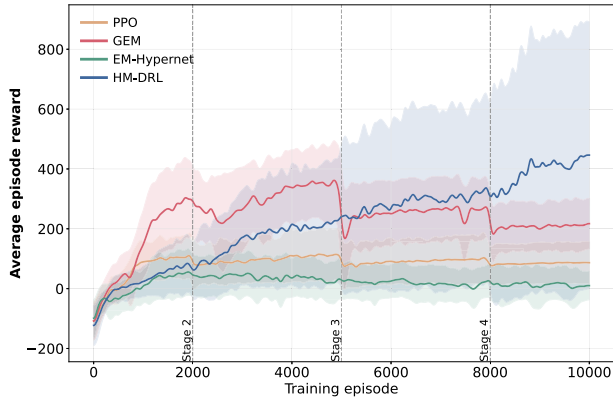
Group	Stage 1	Stage 2	Stage 3	Stage 4	Crisis	Dynamic weather
1	day 5, food 5	day 5, food 3	day 5, food 1	day 10, food 0	0%	Disable
2	day 1, food 5	day 3, food 3	day 5, food 1	day 10, food 0	0%	Disable
3	day 5, food 5	day 5, food 3	day 5, food 1	day 10, food 0	0%	Enable
4	day 5, food 5	day 5, food 3	day 5, food 1	day 10, food 0	10%	Disable



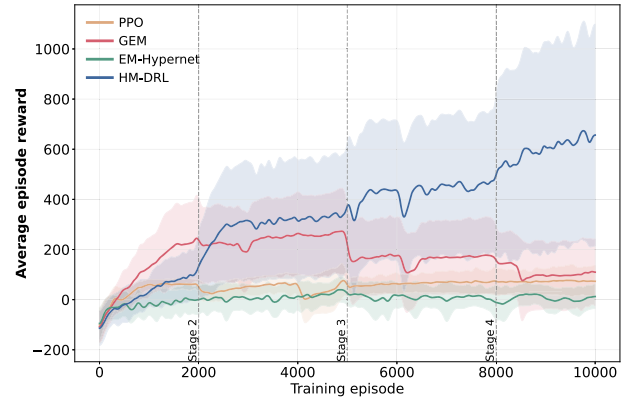
(a) Group 1



(b) Group 2

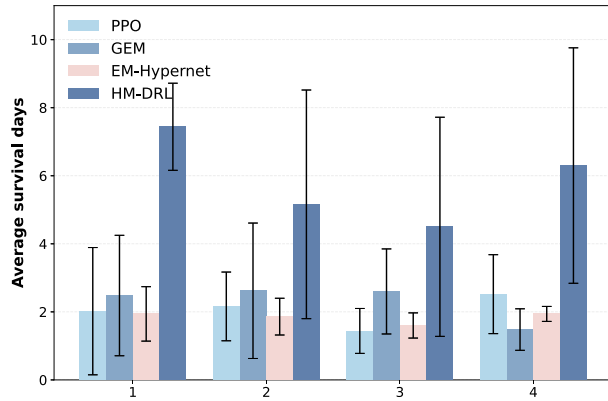


(c) Group 3

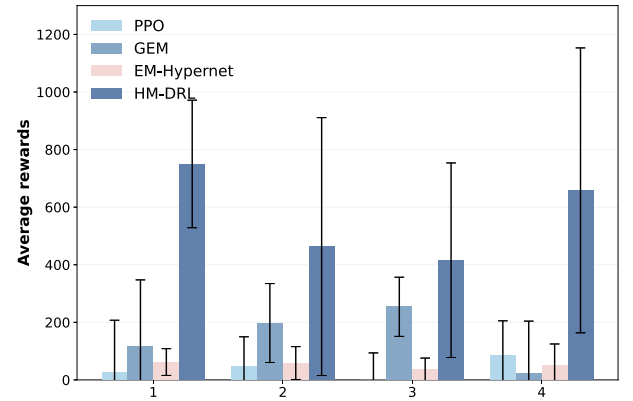


(d) Group 4

Fig. 4. Learning curves of average episodic reward over 10,000 training episodes. The shaded region represents the standard deviation of the performance. Each curve is averaged over five seeds and is smoothed for visual clarity.



(a) Average survival days per agent



(b) Average rewards per agent

Fig. 5. Testing performance comparison across four methods. Each method is trained using five random seeds, and each trained model is evaluated over ten independent test episodes. (a) Average survival days per agent. (b) Average accumulated rewards per agent. Bars indicate the mean performance across seeds, and error bars denote the corresponding standard deviation. HM-DRL generally achieves longer survival duration and higher accumulated rewards than PPO, GEM, and EM-Hypernet, indicating more stable and effective survival behavior.

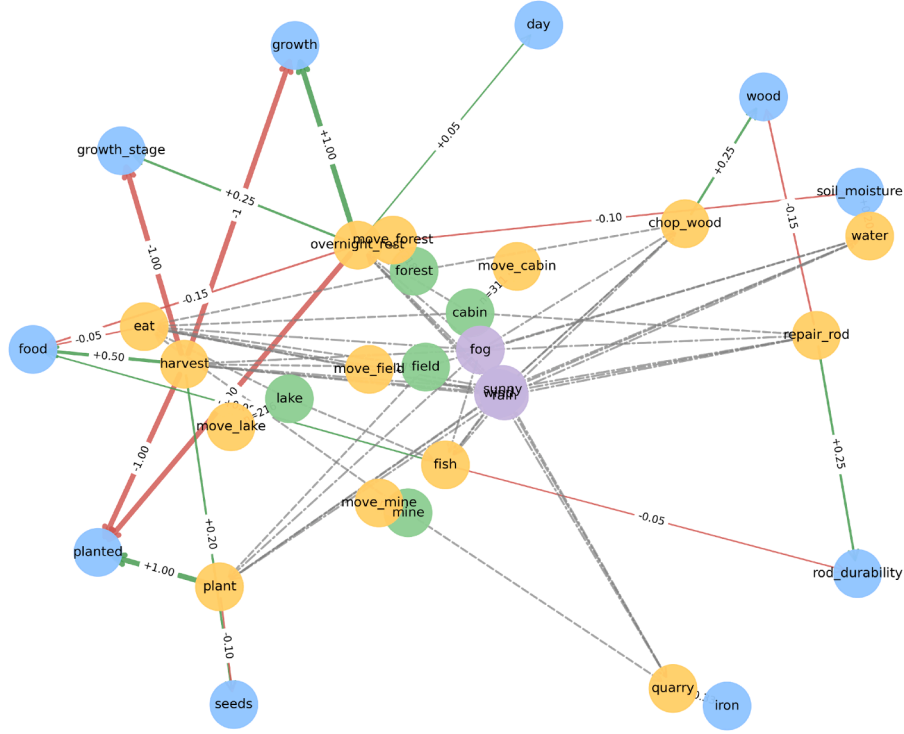


Fig. 6. Visualization of the learned CG. Nodes represent high-level actions (orange), locations (green), and resource dimensions (blue). Purple elements denote contextual factors such as weather, which are not modeled as nodes. Numerical edge labels report the expected effect Δ_e , estimated via state differencing, for each learned precondition-action-effect relation.

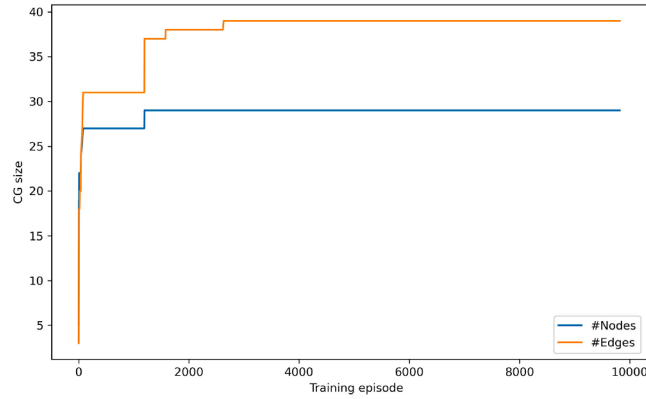


Fig. 7. Evolution of CG size during training. The CG size is measured as the total number of nodes and edges, where nodes represent abstract actions, locations, and resource dimensions, and edges encode learned precondition-action and action-effect relationships.

Table 6

Quantitative results under four task settings. Results are reported as mean $\pm$ standard deviation over five random seeds. Each trained model is evaluated over ten independent test episodes. HM-DRL consistently achieves higher mean performance across all experimental group.

Testing category	Group	PPO	GEM	EM-Hypernet	HM-DRL
Average survival days	1	2.02 $\pm$ 1.87	2.48 $\pm$ 1.77	1.94 $\pm$ 0.8	7.44 $\pm$ 1.28
	2	2.16 $\pm$ 1.01	2.62 $\pm$ 1.99	1.86 $\pm$ 0.54	5.16 $\pm$ 3.36
	3	1.44 $\pm$ 0.67	2.6 $\pm$ 1.25	1.6 $\pm$ 0.37	4.5 $\pm$ 3.22
	4	2.52 $\pm$ 1.16	1.48 $\pm$ 0.61	1.94 $\pm$ 0.22	6.3 $\pm$ 3.46
Average rewards	1	27.3 $\pm$ 179.99	113.84 $\pm$ 231.55	62.06 $\pm$ 46.51	750.1 $\pm$ 221.44
	2	46.4 $\pm$ 103.37	197.56 $\pm$ 137.03	58.56 $\pm$ 57.07	463.1 $\pm$ 447.82
	3	-30.18 $\pm$ 124.04	253.92 $\pm$ 102.79	36.92 $\pm$ 38.93	415.72 $\pm$ 337.86
	4	84.16 $\pm$ 121.11	23.7 $\pm$ 180.5	51.78 $\pm$ 73.02	658.36 $\pm$ 494.75

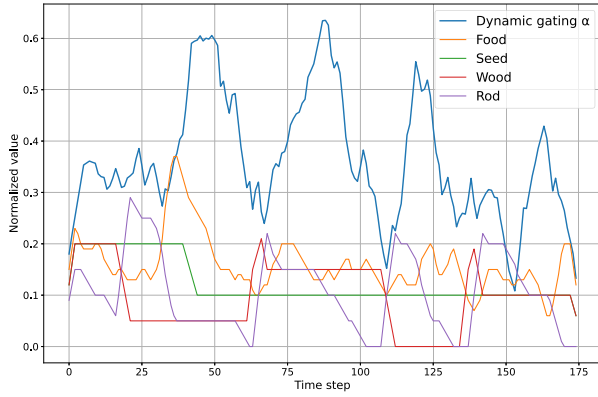
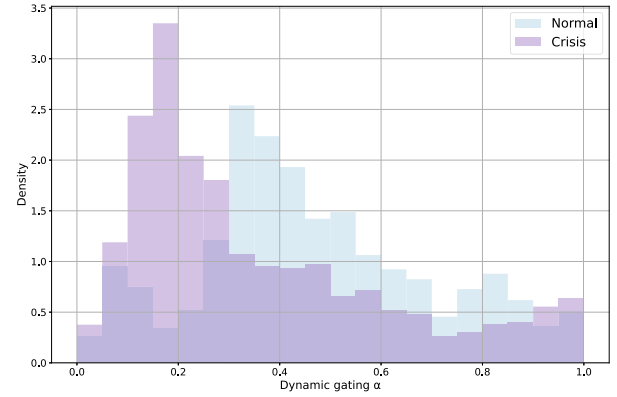
(a) Temporal evolution of α and resources.(b) Distribution of α under normal vs. crisis states.

Fig. 8. Analysis of dynamic gating mechanism. (a) The gate output α varies adaptively with the resource state. (b) The learned gate exhibits different activation patterns in normal and food crisis conditions, demonstrating state-dependent fusion between the policy network and the memory system.

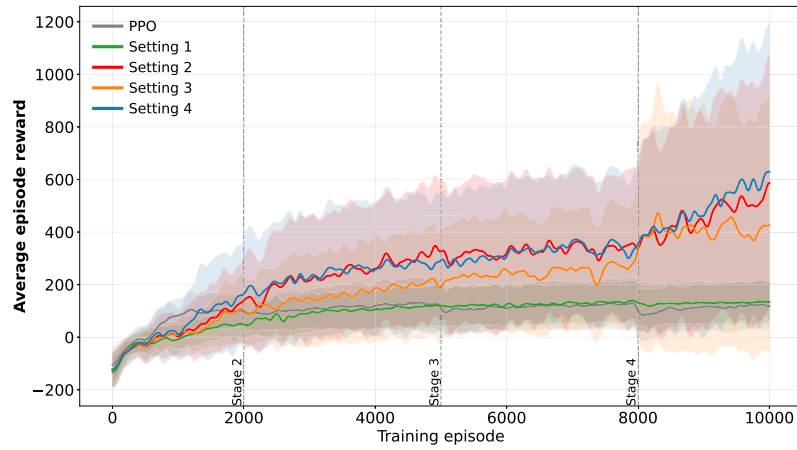


Fig. 9. Ablation study learning curves over 10,000 training episodes (mean $\pm$ std over five seeds; smoothed for visualization). Gray: PPO. Setting 1: w/o (M^a). Setting 2: injected misleading paths ($p = .1$). Setting 3: w/o long-horizon priors. Setting 4: full HM-DRL.

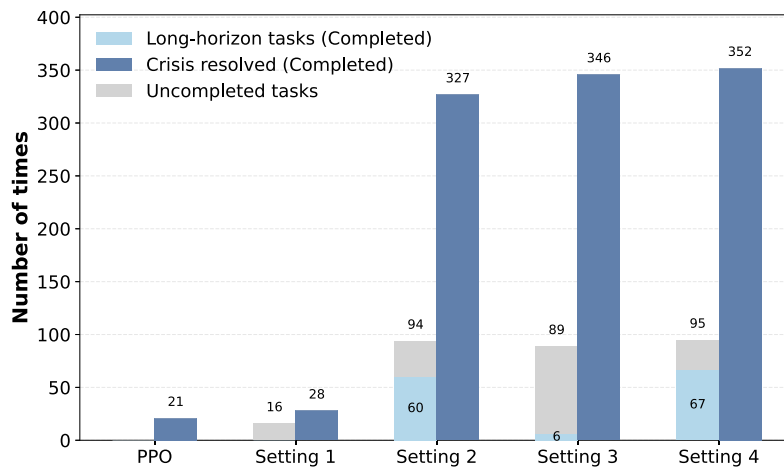


Fig. 10. Results on short-term crisis handling and long-horizon task completion, aggregated over all test episodes. Each model is trained with five random seeds and evaluated over 10 independent test episodes per seed. Dark blue denotes resolved short-term crises, light blue denotes completed long-horizon tasks, and gray denotes unsuccessful cases.

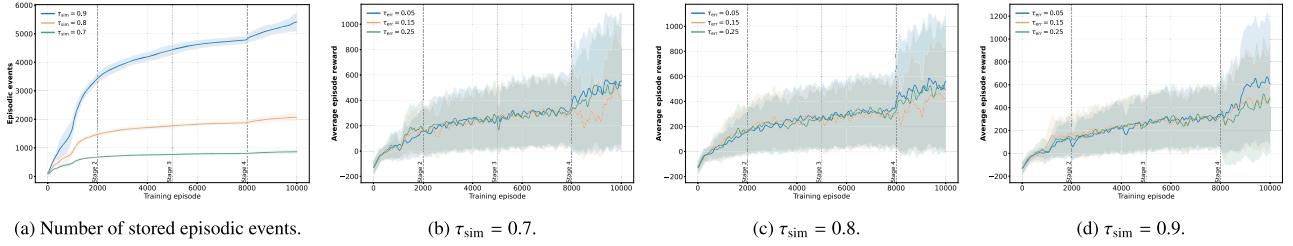


Fig. 11. Sensitivity analysis of two hyperparameters τ_{sim} and τ_{err} . (a) Number of stored episodic events under different similarity thresholds τ_{sim} , which grows substantially as τ_{sim} increases and provides denser coverage of the successful state space. (b)–(d) Training performance under different prediction-error thresholds $\tau_{\text{err}} \in \{0.05, 0.15, 0.25\}$ for each τ_{sim} setting. Curves show the mean and shaded standard deviation across three random seeds.

the low reward associated with the eating action is further depressed by the min operation in twin networks, leading the agent to persistently avoid executing this action. Moreover, GEM relies on previously encountered trajectories stored in the memory buffer; if key survival action sequences are not explored early, subsequent implicit planning cannot generate corresponding virtual trajectories. As a result, GEM agents frequently fall into local optima and fail to perform basic survival planning.

In contrast, agents under the HM-DRL architecture outperformed all baseline methods in both average score and survival duration. During the experiments, we observed a representative phenomenon: when faced with complex tasks and no prior knowledge, agents struggled to discover a reasonable construction order—such as building a crafting table before gathering resources to upgrade equipment—purely through policy convergence. However, once prior knowledge was provided, the agent could rapidly structure this knowledge into a CG, establish causal chains, and subsequently plan survival behaviors based on them. This highlights a general limitation of current RL systems: autonomous discovery of long-horizon task structures from scratch remains challenging without at least sparse supervisory signals.

5.3. Causal graph and dynamic gating analysis

Fig. 6 visualizes the learned CG constructed from canonical rule tuples $\langle \Gamma_a, a, \Delta_a \rangle$ extracted by state differencing. Dashed gray edges from locations to actions encode typical preconditions Γ_a (e.g., *fish* is primarily executed at the lake), while solid edges from actions to resources summarize their average effects Δ_a (green: positive, red: negative; edge thickness scales with $|\Delta_a|$). In addition, the graph incorporates contextual factors such as weather (shown in purple), which are treated as exogenous variables rather than explicit nodes in the CG. These factors modulate action preconditions and effects but do not participate in the structural transitions. Together, the CG makes the slow-learning M^a explicit as a structured knowledge base for long-horizon planning.

As shown in Fig. 7, the size of CG grows rapidly during the early training stage as new precondition–action–effect relations are discovered. After this initial expansion, the curve gradually stabilizes and exhibits little further growth in the mid-to-late training stage, indicating that the M^a converges to a compact and non-redundant set of causal rules rather than expanding indefinitely. This stabilization supports reliable backward reasoning under sparse rewards and crisis disturbances. These observations suggest that the CG captures reusable causal regularities rather than episode-specific correlations.

The gating output α fluctuates dynamically within an episode. When resources remain stable, α tends to increase, indicating stronger reliance on the policy network. In contrast, when food drops sharply, α often decreases, shifting more weight toward memory-based recommendations. This behavior, illustrated in Fig. 8, suggests that the gate acts as a soft regulator that adaptively coordinates learned policies and memory recall, rather than a discrete switch.

5.4. Ablation results

According to the ablation results in Fig. 9, the agent in Setting 1, which lacked the M^a , lost its reasoning ability and achieved only slightly higher average rewards than the pure PPO. In Setting 2, although the agent was affected by misleading paths in the early training phase, policy gradient updates enabled it to progressively calibrate its strategy policy in the later stages, leading to a significant improvement in average rewards. In comparison, Setting 3 achieved lower rewards than the full model, which was attributed to the absence of guidance for long-horizon tasks. This deficiency led to delayed responses to critical tasks, such as harvesting or watering crops, thereby limiting policy learning effectiveness.

As shown in Fig. 10, agents under HM-DRL achieved a long-horizon task completion rate of 70.5% across all evaluation episodes, while the misleading paths group achieved around 63.8%, both substantially outperforming the remaining ablation settings, which yielded nearly 0% success. In particular, Setting 3 attained a markedly lower completion rate of 6.7%, whereas Setting 1 and the PPO baseline failed to complete the task in any evaluation episode. Remarkably, even with 10% erroneous causal recommendations, the agent was able to leverage continuous environmental feedback and policy learning to identify and disregard the misleading information. Ultimately, its performance closely approached that of the complete HM-DRL architecture. These results demonstrate that the policy network does not passively follow path recommendations but instead actively learns and optimizes decision strategies under the combined influence of intrinsic and extrinsic rewards.

5.5. Sensitivity analysis

As shown in Fig. 11, increasing τ_{sim} enlarged the episodic memory, yielding approximately 900, 2100, and 5300 stored events for the three settings, respectively. This higher coverage allowed more states to find nearby episodic neighbors. When combined with a strict prediction-error criterion ($\tau_{\text{err}} = 0.05$), the densified memory at $\tau_{\text{sim}} = 0.9$ produced the best overall performance across all tested configurations.

Error threshold τ_{err} controls how closely current states must match past episodes in order to trigger the r_{predict} . A small threshold enforces high matching precision, ensuring that only truly familiar states yield intrinsic reward signals. This produces a clean and reliable form of guidance that reinforces predictable transitions. In contrast, a large τ_{err} relaxes the matching criterion and increases the likelihood that heterogeneous states are treated as consistent, leading to noisy or misleading intrinsic signals that negatively affect policy learning.

Overall, these findings indicate that a high similarity threshold combined with a strict prediction-matching threshold yields the most stable and effective episodic-guided behavior.

5.6. Discussion

Experimental results show that the proposed HM-DRL architecture, through dynamic coordination of its internal memory layers, consis-

tently outperforms baseline methods under sparse rewards and dynamic crisis disturbances. Compared with the PPO baseline, HM-DRL responds more rapidly to emergent crises, achieves a 70.5% task completion rate in long-horizon tasks, and prolongs survival time by a factor of 3.5. As shown in Group 4 of the ablation study, when random resource disappearance events are introduced, HM-DRL agents employ reasoning mechanisms from M^a to generate goal-directed action sequences, effectively avoiding the oscillatory and ineffective behaviors exhibited by baseline agents. These findings indicate that HM-DRL helps alleviate issues of fragmented memory and limited adaptability in open-world decision-making.

While the above results demonstrate the effectiveness of HM-DRL in handling long-horizon tasks and crisis scenarios, they naturally raise concerns regarding the computational cost of CG-based backward reasoning into the decision-making loop. In practice, the backward reasoning procedure does not perform blind search. At each regression step, only actions whose effects Δ_a contribute to achieving the current subgoal are considered, which substantially reduces the effective branching factor. In our environment, this semantic filtering typically leaves only one or two applicable actions. Moreover, the causal rules $\langle \Gamma_a, a, \Delta_a \rangle$ extracted by the M^a tend to form near-linear chains with limited cross-action dependencies. As a result, even with a maximum search depth of $\beta = 10$, the search remains shallow, and the required executable action is typically identified within a small number of regression steps (often two or three in our environment). Therefore, despite the seemingly large theoretical action space, the reasoning module remains computationally efficient in practice and does not introduce significant variance or instability.

Beyond computational efficiency, it is also important to note the modeling assumptions underlying the CG used in HM-DRL. While the CG provides an interpretable structure for long-horizon reasoning, it is intended as a functional abstraction rather than a statistically identifiable causal model. The extracted rules rely on discrete symbolic state representations; thus the architecture assumes environments where such factors are available or learnable. In more complex or high-dimensional perceptual domains, the number of extracted tuples may grow and increase the complexity of backward reasoning. Scaling HM-DRL to such environments will likely require rule compression, hierarchical abstractions, or relevance-based filtering of causal dependencies.

Furthermore, although the CG captures the effects of environmental factors such as weather, its ability to generate actionable interventions in highly dynamic environments remains limited, highlighting the need for more expressive reasoning mechanisms within the M^a . In future work, we will: 1) enhance the reasoning mechanisms in the M^a to improve its expressiveness and generalization to more complex task structures, while simultaneously reducing the computational overhead of causal reasoning in larger or more dynamic environments, and 2) improve the design of emotion-event associations within the episodic memory layer, envisioning a system where emotional responses to key events can guide decision-making, generate intrinsic motivation, and shape behavioral preferences under similar contexts.

6. Conclusion

In this article, we proposed a hierarchical memory-based deep reinforcement learning (HM-DRL) architecture, a neuroscience-inspired hierarchical memory architecture that integrates perceptual, episodic, and abstract memory systems to enable robust, long-horizon decision-making in non-stationary open-world environments. By dynamically constructing and reusing causal graphs from past experience, the agent achieves effective exploration under sparse rewards, rapid adaptation to unforeseen crises, and sustained knowledge integration-capabilities that remain elusive for conventional deep reinforcement learning approaches.

Our results establish that endowing agents with multi-scale, interacting memory systems is a promising direction toward continual

learning and general-purpose intelligence. Despite current limitations in backward reasoning depth and emotion modeling, the core principles validated here lay a solid foundation for future agents that can autonomously construct knowledge, attribute outcomes, and develop intrinsic behavioral preferences. Moving forward, we aim to close the loop between causal reasoning and affective processing, ultimately enabling emotion and motivation to emerge naturally from the agent's own experience rather than external reward signals alone. This line of research brings us one step closer to artificial systems capable of truly open-ended, lifelong learning.

CRedit authorship contribution statement

Yuhu Cheng: Writing – review & editing, Supervision, Resources, Conceptualization; **Yuequn Zhang:** Writing – original draft, Visualization, Software, Methodology, Investigation, Conceptualization; **C. L. Philip Chen:** Writing – review & editing, Conceptualization; **Xuesong Wang:** Writing – review & editing, Supervision, Resources.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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